Lecture 28

Section 5.7 (Part I)

Differential Equations

Definition: A differential equation is an equation which relates a function to its own derivative.

Example: Consider $y'(t) = ky(t)$, where $y(t)$ is real-valued and k is a real number. From previous calculus courses, we know that $y(t) = Ce^{kt}$, where $C = y(0)$.

Definition: Consider a system of differential equations, namely

$$\begin{aligned} x_1' &= a_{11}x_1 + \cdots + a_{1n}x_n, \\ x_2' &= a_{21}x_1 + \cdots + a_{2n}x_n, \\ &\vdots \\ x_n' &= a_{n1}x_1 + \cdots + a_{nn}x_n, \end{aligned}$$

where $x_1, \ldots, x_n$ are differentiable functions of t, with derivatives $x_1', \ldots, x_n'$, and a_{ij} are constants.

We can write the system of differential equations in the above definition as

$$\mathbf{x}'(t) = A\mathbf{x}(t),$$

where

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}, \qquad \mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \qquad \mathbf{x}'(t) = \begin{pmatrix} x_1'(t) \\ \vdots \\ x_n'(t) \end{pmatrix}.$$

A solution of $\mathbf{x}'(t) = A\mathbf{x}(t)$ is a vector-valued function that satisfies this equation for all t in some interval of real numbers.

Goal: Determine solutions of $\mathbf{x}'(t) = A\mathbf{x}(t)$.

Properties

1. $0 = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$ is a solution.

2. Suppose $\mathbf{u}(t)$ and $\mathbf{v}(t)$ are solutions. Then,

$$A(\mathbf{u}(t) + \mathbf{v}(t)) = A\mathbf{u}(t) + A\mathbf{v}(t) = \mathbf{u}'(t) + \mathbf{v}'(t) = (\mathbf{u}(t) + \mathbf{v}(t))',$$

   implying that $\mathbf{u}(t) + \mathbf{v}(t)$ is a solution.

3. Suppose $\mathbf{u}(t)$ is a solution and c is a real number. Then,

$$A \cdot c\mathbf{u}(t) = c \cdot A\mathbf{u}(t) = c\mathbf{u}'(t) = (c\mathbf{u}(t))',$$

implying $c\mathbf{u}(t)$ is a solution.

Remark: The set of all solutions of $\mathbf{x}' = A\mathbf{x}$ is a subspace of the set of all continuous equations with values in $\mathbb{R}^n$.

Definition: Let A be an $n \times n$ matrix. A fundamental set of solutions is a basis for the set of all solutions of $\mathbf{x}' = A\mathbf{x}$.

Remark: The solution set in the above definition is an n-dimensional vector space of functions.

Suppose we specify a vector $\mathbf{x}_0 \in \mathbb{R}^n$. We want to solve the initial value problem, i.e. construct the unique function $\mathbf{x}$ such that $\mathbf{x}' = A\mathbf{x}$ and $\mathbf{x}(0) = \mathbf{x}_0$.

Example: Suppose

$$\begin{pmatrix} x_1'(t) \\ x_2'(t) \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}.$$

So, $x_1'(t) = -2x_1(t)$ and $x_2'(t) = 4x_2(t)$. Therefore, $x_1(t) = C_1 e^{-2t}$ and $x_2(t) = C_2 e^{4t}$, where C_1, C_2 are constants. So,

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} C_1 e^{-2t} \\ C_2 e^{4t} \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-2t} + C_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{4t}.$$

Therefore, the fundamental set of solutions is

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-2t}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{4t} \right\}.$$

Definition: A system is decoupled if each derivative of a function depends only on the function itself.

Example: The system in the previous example is decoupled, e.g. $x_1'(t)$ is a scalar multiple of $x_1(t)$ (similarly for $x_2'(t)$).

Remark: If A is diagonal, then the corresponding system will be decoupled.

Observation: Consider $\mathbf{x}' = A\mathbf{x}$. A solution is a linear combination of functions of the form $\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$, where λ is a scalar and $\mathbf{v}$ is a fixed nonzero vector. Taking the derivative of both sides of $\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$ with respect to t,

$$\frac{d}{dt}(\mathbf{x}(t)) = \frac{d}{dt}\left(\mathbf{v}e^{\lambda t}\right) \implies \mathbf{x}'(t) = \lambda \mathbf{v} e^{\lambda t}.$$

Applying the matrix A to both sides of $\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$, we get that

$$A\mathbf{x}(t) = A\mathbf{v}e^{\lambda t}.$$

Note that $e^{\lambda t}$ is never equal to 0. So, $A\mathbf{x}(t)$ is equal to $\mathbf{x}'(t)$ if and only if $A\mathbf{v} = \lambda\mathbf{v}$, i.e. if and only if λ is an eigenvalue of A corresponding to the eigenvector $\mathbf{v}$. These solutions are called eigenfunctions of the differential equation.

Example: Suppose

$$A = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}.$$

1. Find a fundamental set of solutions.
2. Solve the initial value problem where $\mathbf{x}' = A\mathbf{x}$ and $\mathbf{x}(0) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Solution: (1) We solve for the characteristic polynomial of A, namely

$$|A - \lambda I_2| = \begin{vmatrix} 3-\lambda & -2 \\ 1 & -\lambda \end{vmatrix} = -\lambda(3-\lambda) + 2 = \lambda^2 - 3\lambda + 2.$$

So, the characteristic equation of A is $\lambda^2 - 3\lambda + 2 = 0$, i.e. $(\lambda - 2)(\lambda - 1) = 0$, implying that the eigenvalues of A are $\lambda = 1, 2$.
Let's fix $\lambda = 1$. We see that

$$A - I_2 = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}.$$

Using row reduction,

$$\begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix} \xrightarrow{r_2 \to 2 \cdot r_2} \begin{pmatrix} 2 & -2 \\ 2 & -2 \end{pmatrix} \xrightarrow{r_2 \to r_2 - r_1} \begin{pmatrix} 2 & -2 \\ 0 & 0 \end{pmatrix} \xrightarrow{r_1 \to \frac{1}{2} r_1} \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}.$$

This implies that $x_1 - x_2 = 0$, i.e. $x_1 = x_2$. So, an eigenvector corresponding to $\lambda = 1$ is of the form

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ x_2 \end{pmatrix} = x_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Now, let's fix $\lambda = 2$. We see that

$$A - 2I_2 = \begin{pmatrix} 1 & -2 \\ 1 & -2 \end{pmatrix}.$$

Using row reduction,

$$\begin{pmatrix} 1 & -2 \\ 1 & -2 \end{pmatrix} \xrightarrow{r_2 \to r_2 - r_1} \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}.$$

This implies that $x_1 - 2x_2 = 0$, i.e. $x_1 = 2x_2$. So, an eigenvector corresponding to $\lambda = 2$ is of the form

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2x_2 \\ x_2 \end{pmatrix} = x_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Hence, the fundamental set of solutions is

$$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t, \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} \right\}.$$

(2) The solutions of $\mathbf{x}' = A\mathbf{x}$ must be of the form

$$\mathbf{x}(t) = C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + C_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t}.$$

If $\mathbf{x}(0) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, then

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \mathbf{x}(0) = C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Putting this information into a matrix and row reducing,

$$\begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & -1 \end{pmatrix} \xrightarrow{r_2 \to r_2 - r_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & -2 \end{pmatrix} \xrightarrow{r_2 \to -1 \cdot r_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{r_1 \to r_1 - 2r_2} \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \end{pmatrix},$$

implying that $C_1 = -3$ and $C_2 = 2$. So,

$$\mathbf{x}(t) = -3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t}.$$